
EpiTwin: Spatiotemporal Graph Transformers for Epileptic sEEG Signal Reconstruction

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Abstract

Stereotactic electroencephalography (sEEG) provides temporally precise intracranial recordings but is inherently constrained by sparse and irregular spatial sampling due to clinical limitations on electrode implantation. Signal reconstruction under this setting aims to infer neural activity at unmonitored locations, potentially expanding the coverage of neural recordings without increasing the number of implanted electrodes. However, most existing sEEG reconstruction methods underutilize the spatial information of electrode contacts in both encoding and modeling, and rely on deterministic objectives that favor average patterns, leading to over-smoothed reconstructions. We propose EpiTwin, a conditional spatial graph transformer for sEEG signal reconstruction, comprising three key components. Hybrid Spatial Positional Encoding (HSPE) constructs explicit spatial identities from electrode coordinates, graph topology, and anatomical priors. Geometry-Functional Biased Attention (GFBA) incorporates geometric distance and data-driven functional similarity biases into attention computation. The adversarial refinement training employs a multi-scale discriminator to counter reconstruction over-smoothing. Experiments on real-world clinical sEEG data demonstrate that EpiTwin consistently achieves lower reconstruction error under electrode series-level masking, outperforming recent foundation models such as LaBraM with a 16.8% relative reduction in RMSE. Furthermore, EpiTwin effectively mitigates spectral over-smoothing and improves reconstruction fidelity.

1. Introduction

Stereotactic electroencephalography (sEEG) is an invasive intracranial recording technique that measures neural activity from deep brain structures via implanted depth electrodes. By directly sampling signals from targeted regions, sEEG provides recordings with higher spatial specificity and reduced contamination from volume conduction and non-neural artifacts compared with scalp electroencephalography (EEG), enabling more accurate observation of seizure-related electrophysiological activity (Bolzan et al., 2024; Parvizi & Kastner, 2018).

Despite these advantages, sEEG measurements are inherently sparse and non-uniform due to surgical safety and anatomical constraints on electrode implantation, providing only a limited and incomplete view of whole-brain activity. Consequently, neural dynamics relevant to seizure generation must often be inferred from partial observations distributed across heterogeneous anatomical locations. Reconstructing neural activity in unobserved regions and modeling interactions between spatially separated brain areas are therefore essential for a more comprehensive understanding of epileptic networks (Jaber et al., 2024; Qi et al., 2020).

In recent years, sEEG signal reconstruction has predominantly been formulated as a masked signal recovery problem. Under this formulation, many recent studies reconstruct masked signals from the remaining unmasked observations. Sequence-based models, including LaBraM (Jiang et al., 2024) and BrainBERT (Wang et al., 2023), leverage generative modeling and self-supervised learning to capture temporal patterns in neural time series. While effective at modeling long-range temporal dependencies, these approaches treat sEEG recordings purely as token sequences and fail to encode the underlying implantation geometry. This missing inductive bias cannot be compensated by increased model capacity under sparse observation regimes.

Another line of work explores graph-based models for sEEG signal reconstruction by representing electrodes as nodes in a graph. However, global spatial configuration and anatomical context are not explicitly modeled in most existing approaches, as their architectures are dominated by local message passing and implicit spatial reasoning. Conse-

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quently, spatial geometry and anatomical priors intrinsic to sEEG recordings remain only partially exploited in current reconstruction frameworks.

In addition to spatial challenges, achieving high-fidelity reconstruction of sEEG signals remains difficult in the spectral domain. Most self-supervised reconstruction methods optimize deterministic objectives such as mean squared error or ℓ_1 loss, which favor average signal patterns (Mathieu et al., 2016; Larsen et al., 2016). Although such objectives can preserve coarse temporal structure, they tend to attenuate fine-grained time-frequency variations and transient spectral features, leading to perceptually over-smoothed reconstructions. Accurately recovering detailed time-frequency structure is therefore critical for faithfully modeling seizure-related neural dynamics, yet remains insufficiently addressed by existing approaches.

To address the challenges of incomplete spatial coverage and limited spectral fidelity in sEEG reconstruction, we propose **EpiTwin**, an end-to-end conditional spatial graph transformer for sEEG signal reconstruction and attention-derived interaction visualization. EpiTwin is designed to explicitly incorporate the physical and anatomical organization of the brain into the reconstruction process, rather than treating spatial relationships as implicit or auxiliary information. The main contributions of this work can be summarized as follows:

- **A unified framework for sEEG signal reconstruction with interaction pattern analysis.** We propose EpiTwin, an end-to-end model that jointly reconstructs sEEG signals from partial observations and supports functional interaction analysis, providing a unified framework for sEEG signal reconstruction under limited electrode coverage.
- **Spatial identity encoding and biased attention.** We introduce a spatial modeling strategy that integrates Hybrid Spatial Positional Encoding (HSPE) with Geometry-Functional Biased Attention (GFBA), endowing each electrode with an informative spatial identity and guiding interactions based on both physical distance and signal-informed relationships.
- **Adversarial refinement for mitigating spectral over-smoothing.** We propose an adversarial refinement strategy with a multi-scale discriminator to mitigate spectral over-smoothing induced by deterministic reconstruction losses, thereby improving the preservation of fine-grained time-frequency structure.

Experimental results on a clinical sEEG dataset demonstrate that EpiTwin achieves lower reconstruction error than existing methods under sparse electrode coverage. Specifically, EpiTwin reduces the root mean squared error (RMSE) by

up to 16.8% compared to strong baseline models on the evaluated benchmark.

2. Related Work

2.1. Generative Masked Modeling for Neural Signals

Generative masked modeling has become a widely adopted paradigm in machine learning for learning representations of complex temporal dynamics (He et al., 2022). Prior studies such as BrainBERT (Wang et al., 2023) and Brant (Zhang et al., 2023) demonstrate its effectiveness under relatively dense or task-specific sEEG settings, while more recent approaches further scale this framework using tokenized sequence modeling (Jiang et al., 2024). Despite their success, these methods primarily treat neural recordings as generic token sequences and focus on temporal structure. They do not explicitly account for the complex spatial and anatomical organization of the brain under sparse sEEG sampling.

2.2. Graph Representation Learning on Brain Networks

Graph neural networks (GNNs) are widely used to model multichannel EEG and sEEG signals by representing electrodes as nodes and learning inter-channel dependencies on brain graphs (Covert et al., 2019; Demir et al., 2021; 2022; Klepl et al., 2022). Most existing methods construct graphs based on data-driven functional similarity. They then apply graph convolution or attention mechanisms to model inter-channel relationships, achieving improved performance in tasks such as seizure detection, motor imagery classification, and neurological disorder diagnosis (Covert et al., 2019; Sun et al., 2021; Demir et al., 2022). Recent studies further incorporate temporal dynamics and hierarchical structures, using spatiotemporal or spectral graph convolutions for seizure prediction (Li et al., 2021), and hierarchical graph diffusion to model epileptic wave propagation in sEEG recordings (Chen et al., 2022). Self-supervised and adaptive graph learning has also been explored to improve robustness to channel variability (Ho & Armanfard, 2023). However, in most existing EEG graph-based approaches, graph structure is primarily used for feature aggregation or representation learning, rather than serving as an explicit spatial inductive bias for inferring unobserved signals.

2.3. High-Fidelity Spectrogram Reconstruction

High-fidelity reconstruction is critical for preserving fine-grained time-frequency structures that are easily suppressed by spectral over-smoothing. Standard self-supervised frameworks such as Ti-MAE (Li et al., 2023) and SimMTM (Dong et al., 2023) typically optimize deterministic losses (e.g., MSE or L1), which can lead to spectrogram over-smoothing and obscure fine-grained and transient time-frequency details. While generative approaches from audio synthe-

sis (Lee et al., 2023; Kong et al., 2020) and diffusion models (Alcaraz & Strodthoff, 2023) offer improved texture recovery, they often incur high computational costs or risk stochastic hallucinations, highlighting the need for efficient, structure-preserving refinement strategies in practical sEEG analysis pipelines.

3. Methodology

We propose **EpiTwin**, an end-to-end conditional spatiotemporal graph transformer model (Ying et al., 2021; He et al., 2022) for high-fidelity sEEG signal reconstruction in epileptic brain networks (Rubinov & Sporns, 2010). Figure 1 provides an overview of the model architecture.

Formally, we model an sEEG implantation as a brain electrode graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A}_0)$, where $\mathcal{V} = \{v_i\}_{i=1}^N$ denotes the set of N electrode nodes, $\mathbf{A}_0 \in \{0, 1\}^{N \times N}$ is constructed as a symmetric k -nearest neighbor graph with $k = 8$ in the 3D Euclidean space of electrode coordinates $\{\mathbf{c}_i\}$, and $\mathcal{E}$ denotes the corresponding edge set induced by the nonzero entries of $\mathbf{A}_0$. We represent each sEEG electrode as a node in the graph. Each node v_i is associated with a log-amplitude spectrogram $\mathbf{s}_i \in \mathbb{R}^{F \times M}$, where F and M denote the numbers of frequency bins and time frames, respectively, spatial coordinates $\mathbf{c}_i \in \mathbb{R}^3$, and anatomical attributes including a region label r_i and tissue composition metrics $\mathbf{m}_i \in \mathbb{R}^2$ (gray-matter and white-matter ratios).

The primary objective of the model is masked sEEG signal reconstruction, which infers the features $\hat{\mathbf{S}}_{\mathcal{V}_T}$ of target nodes $\mathcal{V}_T$ from the features $\mathbf{S}_{\mathcal{V}_O}$ of observed nodes $\mathcal{V}_O$ under sparse spatial sampling. In addition to this primary objective, the attention mechanisms learned for reconstruction naturally induce an asymmetric, attention-derived interaction pattern among the nodes. We expose this pattern as an attention-derived interaction matrix $\hat{\mathbf{A}}_P$, which reflects reconstruction-driven information flow and serves as an interpretable by-product rather than a separately supervised prediction target. Accordingly, the model can be expressed as a mapping function $f_\theta : (\mathbf{S}_{\mathcal{V}_O}, \mathcal{G}) \rightarrow (\hat{\mathbf{S}}_{\mathcal{V}_T}, \hat{\mathbf{A}}_P)$, where $\hat{\mathbf{A}}_P$ is derived from the learned attention weights without explicit optimization objectives.

The EpiTwin architecture comprises three core components: a Heterogeneous Spatiotemporal Information Encoder, a Geometry-Functional Biased Graph Transformer Backbone, and a GAN-based High-Fidelity Refinement Module.

3.1. Heterogeneous Spatiotemporal Information Encoder

The model input comprises two primary heterogeneous information streams: time-varying dynamic features and time-invariant static priors.

Dynamic Feature Extraction. To process continuous, arbitrary-length sEEG signals, we adopt a patch-based segmentation strategy inspired by the Vision Transformer (Dosovitskiy et al., 2021). Given a continuous signal $\mathbf{X} \in \mathbb{R}^{N \times T}$, we segment it along the time dimension into K fixed-length windows of size W , each serving as an independent sample: $\mathbf{x}_k = \mathbf{X}[:, (k-1)W : kW]$, $k \in \{1, \dots, K\}$. This operation transforms the continuous sequence into a batch of fixed-size inputs suitable for downstream feature extraction.

We apply the Short-Time Fourier Transform (STFT) independently to each channel in each patch $\mathbf{x}_k$, yielding a complex-valued spectrogram $\mathbf{Z}_k = \text{STFT}(\mathbf{x}_k) \in \mathbb{C}^{N \times F \times M}$. We use the log-amplitude spectrogram $\mathbf{S}_k = \log(|\mathbf{Z}_k|) \in \mathbb{R}^{N \times F \times M}$ as the dynamic input feature, which is well suited for non-stationary sEEG signals and provides more stable optimization than raw waveforms (Wu et al., 2023). Detailed STFT parameterization is provided in Appendix A.

The tensor $\mathbf{S}_k$ is fed into a node-wise feature encoder, implemented as a lightweight convolutional network, which processes the $F \times M$ spectral map associated with each node and projects it into a d -dimensional latent vector, producing the dynamic feature matrix $\mathbf{H}^{(0)} \in \mathbb{R}^{N \times d}$, where d is the consistent hidden dimension used throughout the model.

Static Priors. Although $\mathbf{H}^{(0)}$ captures the signal’s content, relying solely on signal-driven features presents significant limitations: this stream lacks awareness of the electrodes’ physical spatial locations and fails to leverage crucial biological prior knowledge. In epilepsy analysis, the electrodes’ time-invariant spatial and anatomical attributes are essential biological priors that guide the model’s understanding of pathogenesis and information propagation pathways.

To bridge this gap and equip the model with spatial awareness and clinical context, we propose the **Hybrid Spatial Positional Encoding (HSPE)** mechanism. HSPE integrates these complementary, time-invariant static priors into a single embedding matrix $\mathbf{E} \in \mathbb{R}^{N \times d}$, which is then injected into the dynamic feature stream via residual addition. This mechanism allows subsequent graph attention layers to compute node interaction weights based simultaneously on signal similarity (content) and spatial/biological relationships (identity).

Traditional sequential positional encodings are limited to capturing channel order. This ordering is often arbitrary and lacks physical meaning for sEEG, as it discards the electrodes’ three-dimensional spatial arrangement. Given that sEEG sensors are arrayed on a 3D manifold, we represent the physical position of each electrode i by its coordinates $\mathbf{c}_i = (x_i, y_i, z_i)$. To embed spatial awareness, we utilize

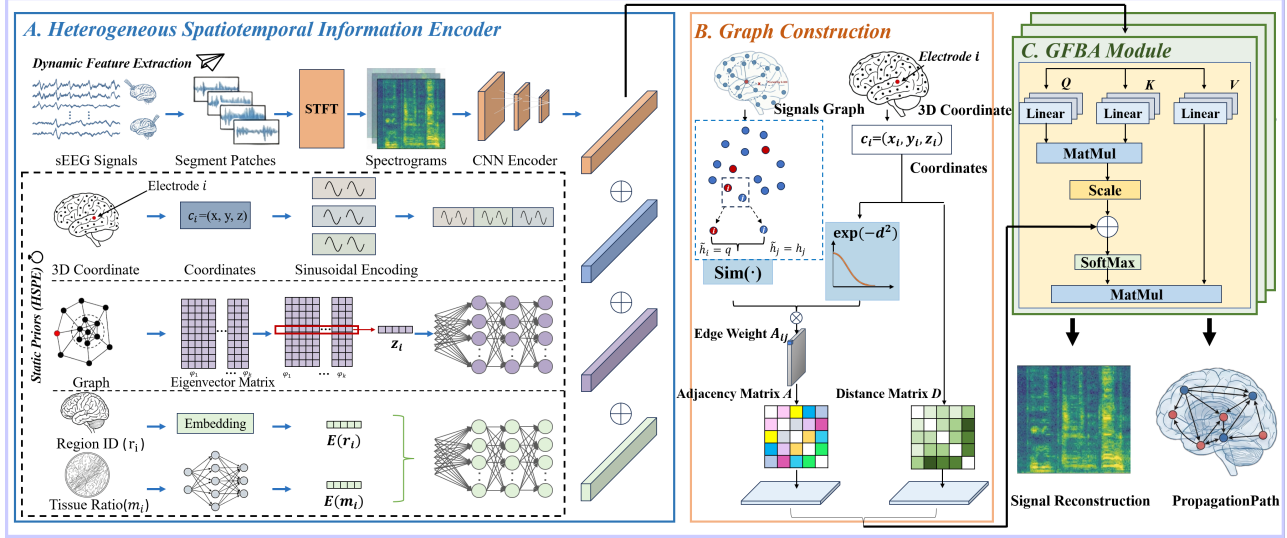


Figure 1. **Architecture of EpiTwin.** An end-to-end conditional spatiotemporal graph transformer for masked sEEG signal reconstruction.

sinusoidal functions to encode each coordinate dimension independently and then **concatenate** them.

Specifically, we divide the total feature dimension d into three sub-segments. For a specific coordinate axis $j \in \{x, y, z\}$ and a sub-dimension index k , the encoding $\mathbf{p}_{i,j}$ is defined as:

$$\mathbf{p}_{i,j}[k] = \begin{cases} \sin\left(\frac{\mathbf{c}_i[j]}{\Omega^{2k/(d/3)}}\right), & \text{if } k \text{ is even,} \\ \cos\left(\frac{\mathbf{c}_i[j]}{\Omega^{2k/(d/3)}}\right), & \text{if } k \text{ is odd.} \end{cases} \quad (1)$$

The constant Ω controls the frequency spectrum of the sinusoidal positional encoding. Its specific value and implementation details are provided in Appendix B.1. The final 3D positional encoding is obtained by $\mathbf{p}_i = \mathbf{p}_{i,x} \oplus \mathbf{p}_{i,y} \oplus \mathbf{p}_{i,z} \in \mathbb{R}^d$, where $\oplus$ denotes the concatenation operation.

To overcome the expressiveness limitation of 1-Weisfeiler-Leman (1-WL) message passing and encode global graph topology, we introduce **Topological Structure Encoding** ($\mathbf{v}_i$) (Xu et al., 2019). A common approach to inject global structural information beyond local message passing is through spectral positional encodings derived from the graph Laplacian. Specifically, we construct the graph $\mathcal{G}$ based on electrode geometry, with adjacency matrix $\mathbf{A}_0$ and degree matrix $\mathbf{D}$, and define the normalized Laplacian as $\mathbf{L}_G = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A}_0 \mathbf{D}^{-1/2}$. We perform an eigendecomposition of $\mathbf{L}_G$ and select the first J smallest non-zero eigenvectors $\{\phi_1, \dots, \phi_J\}$, where $J \ll N$ is a tunable hyperparameter controlling the resolution of the global topology. For each electrode i , the resulting spectral descriptor $\mathbf{z}_i = [\phi_1[i], \dots, \phi_J[i]]^\top$ is projected into the latent space via an MLP, yielding a d -dimensional embedding $\mathbf{v}_i = \text{MLP}_t(\mathbf{z}_i) \in \mathbb{R}^d$. This geometry-based Laplacian is

fixed throughout training and inference, ensuring that the encoding captures stable spatial identity rather than transient functional correlations.

Simultaneously, to infuse high-value biological and clinical knowledge, we construct an **Anatomical Encoding** ($\mathbf{u}_i$). This encoding integrates the macroscopic anatomical brain region ID r_i , mapped via an embedding layer $\mathbf{E}_R(r_i)$, and the tissue ratio at the electrode tip $\mathbf{m}_i = [\text{GM}\%, \text{WM}\%]$, encoded via a lightweight MLP $\mathbf{E}_M(\mathbf{m}_i)$. The final anatomical encoding is given by $\mathbf{u}_i = \text{MLP}_p([\mathbf{E}_R(r_i); \mathbf{E}_M(\mathbf{m}_i)]) \in \mathbb{R}^d$.

By fusing these three complementary encodings, we obtain the final static prior for each node as $\mathbf{e}_i = \mathbf{p}_i + \mathbf{v}_i + \mathbf{u}_i$. Stacking all node-wise priors yields the matrix $\mathbf{E} = [\mathbf{e}_1; \dots; \mathbf{e}_N] \in \mathbb{R}^{N \times d}$, which is residually injected into the dynamic features as $\mathbf{Z}_{\text{final}} = \mathbf{H}^{(0)} + \mathbf{E}$. Implementation details of HSPE, including hyperparameter choices and MLP architectures, are provided in Appendix B.1.

3.2. Spatial Modeling with Geometry-Functional Biased Graph Transformer

A purely geometry-based connectivity pattern (e.g., k -NN) is sparse and physically plausible but ignores functional relationships, whereas a purely function-driven pattern is computationally expensive and susceptible to noise, often introducing implausible long-range links. Therefore, we construct a sparse graph that integrates spatial constraints with signal-driven functional similarity, formally represented as $\mathcal{G}$ with adjacency matrix $\mathbf{A}$ to guide information propagation in the graph transformer.

Masked Graph Construction. To prevent information leakage and ensure strict consistency between training and

inference, we employ a masked graph construction strategy. Before topology inference, the features of all target nodes ($i \in \mathcal{V}_T$) are replaced by a shared learnable query vector $\mathbf{q}$, while observed nodes ($i \in \mathcal{V}_O$) retain their extracted features $\mathbf{h}_i^{(0)}$. Let $\tilde{\mathbf{h}}_i$ denote the feature used for topology inference: $\tilde{\mathbf{h}}_i = \mathbf{h}_i^{(0)}$ for $i \in \mathcal{V}_O$, and $\tilde{\mathbf{h}}_i = \mathbf{q}$ for $i \in \mathcal{V}_T$. Based on these masked features, we define the adjacency matrix $\mathbf{A}$ by jointly considering data-driven functional similarity and spatial proximity:

$$\mathbf{A}_{ij} = \begin{cases} \text{Sim}(\tilde{\mathbf{h}}_i, \tilde{\mathbf{h}}_j) \exp\left(-\frac{\|\mathbf{c}_i - \mathbf{c}_j\|^2}{2\sigma^2}\right), & j \in \mathcal{N}_k(i), \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where $\text{Sim}(\cdot, \cdot)$ denotes cosine similarity and $\mathcal{N}_k(i)$ is the set of k nearest spatial neighbors of node i .

This formulation serves a dual purpose: for observed nodes, edges reflect both signal synchrony and spatial locality, while for target nodes ($\tilde{\mathbf{h}}_i = \mathbf{q}$), the functional term reduces to a query-conditioned similarity with neighboring signals under strong spatial regularization. This design prevents information leakage from target nodes while preserving expressive, spatially grounded functional bias. Implementation details of the masked graph construction, including the choice of similarity function, the spatial decay parameter σ and neighborhood size, are provided in Appendix B.3.

Geometry-Functional Biased Attention (GFBA). The graph transformer backbone refines node representations over the constructed graph $\mathcal{G}$. Standard self-attention is permutation-invariant and therefore agnostic to spatial topology. To explicitly incorporate brain-network structure, we propose **Geometry-Functional Biased Attention (GFBA)**, which injects geometric and functional priors directly into the attention computation.

At Transformer layer l , node representations $\mathbf{Z}^{(l-1)}$ are projected to queries $\mathbf{Q}^{(l)}$, keys $\mathbf{K}^{(l)}$, and values $\mathbf{V}^{(l)}$. In parallel, two bias terms are computed: a geometric bias $\mathbf{B}_G^{(l)}$ based on pairwise Euclidean distances, and a functional bias $\mathbf{B}_F^{(l)}$ based on the adjacency matrix $\mathbf{A}$. For attention head h , the biased attention logits are given by:

$$\mathbf{S}_h^{(l)} = \frac{\mathbf{Q}_h^{(l)} (\mathbf{K}_h^{(l)})^\top}{\sqrt{d_h}} + \mathbf{B}_{G,h}^{(l)} + \mathbf{B}_{F,h}^{(l)}, \quad (3)$$

where $d_h = d/H$ denotes the feature dimension of each attention head, and H is the number of heads.

The attention weights are obtained via $\alpha_h^{(l)} = \text{Softmax}(\mathbf{S}_h^{(l)})$ and used to aggregate value vectors. Each Transformer block follows a standard Pre-LN residual architecture:

$$\begin{aligned} \tilde{\mathbf{Z}}^{(l)} &= \mathbf{Z}^{(l-1)} + \text{GFBA}^{(l)}(\text{LN}(\mathbf{Z}^{(l-1)})), \\ \mathbf{Z}^{(l)} &= \tilde{\mathbf{Z}}^{(l)} + \text{FFN}^{(l)}(\text{LN}(\tilde{\mathbf{Z}}^{(l)})). \end{aligned} \quad (4)$$

In this formulation, $\text{GFBA}^{(l)}(\cdot)$ denotes the Geometry-Functional Biased multi-head attention module at layer l , $\text{LN}(\cdot)$ represents layer normalization, and $\text{FFN}^{(l)}(\cdot)$ is a position-wise feed-forward network, following a standard Pre-LN Transformer architecture. Implementation details of Geometry-Functional Biased Attention, including the construction of geometric and functional bias terms and the Transformer configuration, are provided in Appendix B.3.

Asymmetric Masking for Reconstruction. To steer the Transformer toward the reconstruction task, we apply asymmetric masking at the input. For observed nodes, the input is $\mathbf{h}_i^{(0)} + \mathbf{e}_i$, while for target nodes, the spectral features are replaced by the shared query vector $\mathbf{q}$ and only spatial priors are retained:

$$\mathbf{z}_i^{(0)} = \begin{cases} \mathbf{h}_i^{(0)} + \mathbf{e}_i, & i \in \mathcal{V}_O, \\ \mathbf{q} + \mathbf{e}_i, & i \in \mathcal{V}_T. \end{cases} \quad (5)$$

We derive an interaction matrix $\hat{\mathbf{A}}_P$ by averaging the attention weights from the final Transformer block across heads, which is used solely for post-hoc analysis and qualitative visualization.

$$\hat{\mathbf{A}}_P = \frac{1}{H} \sum_{h=1}^H \text{Softmax}\left(\frac{\mathbf{Q}_h^{(L)} (\mathbf{K}_h^{(L)})^\top}{\sqrt{d_h}} + \mathbf{B}_{G,h}^{(L)} + \mathbf{B}_{F,h}^{(L)}\right). \quad (6)$$

Frequency-Exclusive Progressive Decoding. To reconstruct high-resolution target spectra, we employ a frequency-exclusive progressive decoder. The latent embedding $\mathbf{z}^{(L)}$ is reshaped into an initial feature map with full temporal resolution and is progressively upsampled only along the frequency dimension using anisotropic transposed convolutions, as preserving temporal alignment is critical for capturing transient neural events while spectral detail can be gradually refined. A final 1×1 convolution produces the reconstructed spectrum $\hat{\mathbf{s}}_i$. Architectural details are provided in Appendix B.4.

3.3. Adversarial-Refined Masked Signal Reconstruction

We train the graph transformer framework defined in Section 3.2 to perform sEEG spectrogram reconstruction using a decoupled, two-stage training strategy, as illustrated in Figure 2.

Phase 1: L1-based Initialization. In the first phase, we pretrain the model (the Generator, G) on the Masked Signal Reconstruction (MSR) task (He et al., 2022), reconstructing $\mathcal{V}_T$ from $\mathcal{V}_O$ using only an L1 reconstruction loss: $\mathcal{L}_r = \mathcal{L}_{\text{MAE}}$. This stage provides a stable initialization that encourages the model to capture the correct coarse-grained spectral structure.

Phase 2: Adversarial Refinement. To address the over-

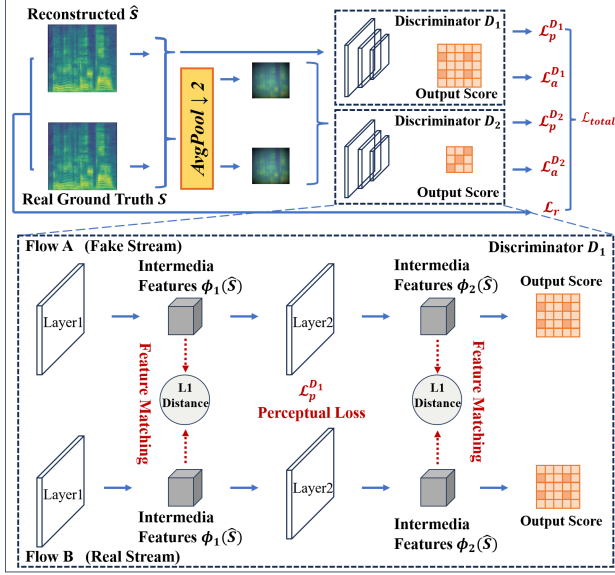


Figure 2. **Adversarial Refinement Training Process of EpiTwin.** The EpiTwin Generator is jointly trained with a multi-scale discriminator to refine reconstructed sEEG signals and mitigate over-smoothing.

smoothing induced by pure L1 optimization, we introduce an adversarial refinement phase based on a **Multi-Scale Discriminator** $D = \{D_1, D_2\}$ with a PatchGAN architecture (Isola et al., 2017). The discriminator is optimized via a multi-scale hinge loss $\mathcal{L}_D$ (Lim & Ye, 2017) to distinguish real signals (x) from reconstructed signals ($\hat{x}$):

$$\mathcal{L}_D = \sum_{k=1}^2 \left(\mathbb{E}_x [\max(0, 1 - D_k(x))] + \mathbb{E}_{\hat{x}} [\max(0, 1 + D_k(\hat{x}))] \right), \quad (7)$$

where D_k denotes the discriminator at the k -th scale. This objective is stabilized by spectral normalization (Miyato et al., 2018).

Conversely, the Generator (G , i.e., the EpiTwin model) aims to fool the discriminator while preserving content structure. Its objective function transitions to a composite loss $\mathcal{L}_G$:

$$\mathcal{L}_G = \lambda_r \mathcal{L}_r + \lambda_a \mathcal{L}_a + \lambda_p \mathcal{L}_p. \quad (8)$$

This formulation combines the L1 reconstruction loss ($\mathcal{L}_r$), the adversarial loss ($\mathcal{L}_a = -\sum_k \mathbb{E}_{\hat{x}} [D_k(\hat{x})]$), and the perceptual loss ($\mathcal{L}_p$), computed from intermediate discriminator features.

Crucially, we down-weight the L1 fidelity term ($\lambda_r = 0.5$), allowing the adversarial and perceptual components, weighted by $\lambda_a = 1.0$ and $\lambda_p = 0.5$, to dominate gradient flow, while maintaining content fidelity via the reconstruction term and the masked setting. This decoupled

initialize-then-refine strategy is trained in an alternating fashion ($N_D = 1, N_G = 1$) and is critical for achieving high-fidelity reconstruction. Implementation details are provided in Appendix B.5.

4. Experiments

4.1. Experimental Setup

Dataset and Preprocessing. We evaluate EpiTwin on a clinical stereotactic EEG (sEEG) cohort of drug-resistant epilepsy patients. Signals are re-referenced using a bipolar montage, sampled at 200 Hz, and band-pass filtered to 0.5-50 Hz. We compute log-amplitude spectrograms by applying STFT to 10-second segments with a 256-point Hann window, resulting in inputs of size 64×16 (frequency $\times$ time). The number of bipolar channels varies across patients (81-124) due to heterogeneous implantation coverage. For each bipolar channel, we provide static metadata (3D coordinates, tissue composition, and anatomical region labels), which are used by all EpiTwin variants and kept fixed across experiments. Additional dataset details are provided in Appendix E.

Evaluation Protocol. We evaluate reconstruction under increasing inter-subject variability using three settings from the same cohort: (i) patient-specific (1 patient), (ii) limited cross-subject (2 patients), and (iii) heterogeneous multi-subject (5 patients). Unless otherwise stated, the single-patient setting serves as the primary benchmark, while the multi-subject settings assess robustness to heterogeneous electrode geometries and neural dynamics. Across all settings, samples are split into training/validation/test sets (60/20/20) at the sample level. For multi-subject experiments, subject-stratified sampling is used, and bipolar channels are aligned to a common dimensionality to control for variability in channel count. A *series-aware* masking protocol is applied during both training and inference, where all bipolar channels from the same electrode shaft are masked simultaneously. All preprocessing, masking, and splitting strategies are shared across methods, with details provided in Appendix E.

Baselines and Metrics. We compare EpiTwin with three conventional neural baselines (Bi-LSTM (Graves & Schmidhuber, 2005), EEGNet (Lawhern et al., 2018), DGCNN (Wang et al., 2019)) and several recent brain foundation models (Brant (Zhang et al., 2023), BrainBERT (Wang et al., 2023), LaBraM (Jiang et al., 2024)). To ensure fair comparison, all baselines are retrained from scratch on the same STFT inputs and under the same series-aware masking and split protocol. We report log-spectrogram RMSE and Pearson correlation coefficient (PCC), and frequency-domain log-spectral distance (LSD); we further compute High-LSD to emphasize sensitivity

Table 1. Quantitative Comparison with State-of-the-Art Methods. We report the mean $\pm$ standard deviation across 5 independent runs. EpiTwin outperforms all baselines, particularly in spectral metrics (LSD). Best results are **bolded**, second best are underlined.

Method	Log-Spectrogram		Freq-Domain	
	RMSE $\downarrow$	PCC $\uparrow$	LSD $\downarrow$	High-LSD $\downarrow$
<i>Traditional Models</i>				
Bi-LSTM (Graves & Schmidhuber, 2005)	0.6160 $\pm$ 0.0021	<u>0.9920</u> $\pm$ 0.0005	0.5660 $\pm$ 0.0032	0.5700 $\pm$ 0.0041
EEGNet (Lawhern et al., 2018)	1.0620 $\pm$ 0.0150	0.9910 $\pm$ 0.0008	0.9440 $\pm$ 0.0055	0.9200 $\pm$ 0.0062
DGCNN (Wang et al., 2019)	0.9050 $\pm$ 0.0089	0.9910 $\pm$ 0.0007	0.8220 $\pm$ 0.0048	0.8100 $\pm$ 0.0053
<i>Brain Foundation Models</i>				
Brant (Zhang et al., 2023)	1.7860 $\pm$ 0.0210	0.9340 $\pm$ 0.0015	1.4390 $\pm$ 0.0120	1.7070 $\pm$ 0.0155
BrainBERT (Wang et al., 2023)	1.3240 $\pm$ 0.0180	0.9740 $\pm$ 0.0011	1.1330 $\pm$ 0.0095	1.2110 $\pm$ 0.0110
LaBraM (Jiang et al., 2024)	<u>0.5910</u> $\pm$ 0.0018	<u>0.9920</u> $\pm$ 0.0004	<u>0.5410</u> $\pm$ 0.0025	<u>0.5250</u> $\pm$ 0.0030
EpiTwin (Ours)	0.4920 $\pm$ 0.0012	0.9940 $\pm$ 0.0003	0.4630 $\pm$ 0.0015	0.4490 $\pm$ 0.0018

to high-frequency over-smoothing. Metric definitions and baseline configurations are provided in Appendix C.

4.2. Main Results

We first evaluate EpiTwin under the single-patient setting, which serves as the primary benchmark for assessing reconstruction performance without inter-subject variability. All results reported in this section are obtained using recordings from a single patient under the series-aware masking protocol. As shown in Table 1, EpiTwin consistently outperforms all baseline methods across both log-spectrogram and frequency-domain metrics. In terms of overall reconstruction accuracy, EpiTwin achieves the lowest RMSE of 0.492, substantially improving upon the strongest foundation model baseline, LaBraM (RMSE 0.591), corresponding to a 16.8% relative reduction in RMSE. This performance gap indicates that explicitly encoding spatial and anatomical inductive bias is more effective than generic token-based modeling for sEEG reconstruction under sparse and irregular electrode coverage.

In addition to reducing reconstruction error, EpiTwin attains the highest waveform similarity, achieving a PCC of 0.994, compared to 0.992 for the best-performing baselines. Notably, several deterministic models achieve competitive PCC values but suffer from pronounced spectral distortions, highlighting that high temporal correlation alone is insufficient to ensure faithful signal reconstruction. The advantage of EpiTwin is particularly evident in the frequency domain. As reported in Table 1, EpiTwin achieves the lowest log-spectral distance, reducing LSD from 0.541 (LaBraM) to 0.463, and further lowers High-LSD from 0.525 to 0.449. These improvements indicate a substantial reduction in spectral over-smoothing, especially in high-frequency bands that are critical for preserving fine-grained neural dynamics. Importantly, these gains in spectral fidelity do not come at the cost of degraded log-spectrogram reconstruction quality. EpiTwin achieves a favorable balance between log-

spectrogram fidelity (RMSE/PCC) and frequency-domain accuracy (LSD/High-LSD), resulting in the most balanced overall performance among all compared methods.

4.3. Ablation Studies

We perform a component-wise ablation study to assess the contribution of each architectural design under the single-patient setting (Table 2). Replacing the proposed hierarchical spatial encoding with any single-source variant (geometric-only, topological-only, or positional-only) consistently degrades performance, with RMSE increasing from 0.492 to approximately 0.51 and corresponding increases in LSD and High-LSD. Similarly, substituting the proposed geometry-functional biased attention with generic attention mechanisms results in inferior reconstruction quality, particularly in the frequency domain (e.g., LSD increases from 0.463 to 0.489). These results indicate that both heterogeneous spatial encoding and biased attention are essential for accurate sEEG reconstruction.

We further examine the effect of adversarial refinement. Removing adversarial training primarily impacts spectral fidelity rather than waveform-level accuracy. As illustrated in Fig. 3, models trained solely with $\mathcal{L}_1$ loss exhibit pronounced spectral over-smoothing, while adversarial refinement recovers sharper time-frequency structures, especially in high-frequency bands. This observation is consistent with the reduced LSD and High-LSD values achieved by the full model, indicating that adversarial refinement complements deterministic objectives by mitigating spectral over-smoothing. Additional qualitative visualizations and extended quantitative results are provided in Appendix D.

4.4. Robustness under Different Dataset Scales

Beyond the single-patient setting, we evaluate how reconstruction performance evolves as inter-subject variability is introduced. Specifically, we consider two expanded set-

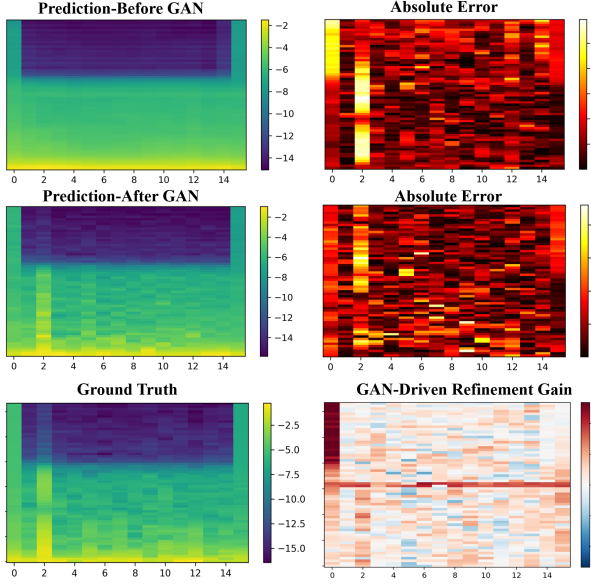


Figure 3. **Qualitative Analysis of Adversarial Refinement.** Adversarial refinement preserves sharper temporal and spectral details compared to reconstruction without adversarial training.

Table 2. **Component-wise Ablation Results.** Bold indicates the best performance of the full EpiTwin model.

Method / Variant	Log-Spectrogram		Freq-Domain	
	RMSE ↓	PCC ↑	LSD ↓	High-LSD ↓
Encoding Strategies				
Positional Only	0.5096	0.9932	0.4807	0.4596
Topological Only	0.5077	0.9933	0.4790	0.4612
Geometric Only	0.5107	0.9932	0.4815	0.4602
Attention Mechanisms				
Vanilla Attention	0.5220	0.9929	0.4894	0.4711
Graph k-NN	0.5191	0.9930	0.4877	0.4698
EpiTwin (Full)	0.4920	0.9940	0.4630	0.4490

tings that include recordings from two and five subjects, respectively, following the dataset construction and splitting protocol described in Section 4.1. In both cases, subject-stratified splits are adopted to ensure that training, validation, and test sets all contain samples from each subject.

As summarized in Table 3 and Table 4, reconstruction performance degrades for all methods as the number of subjects increases, reflecting the increased difficulty of modeling heterogeneous implantation geometries and neural dynamics. Nevertheless, EpiTwin achieves the best RMSE and PCC in both the two-subject and five-subject settings. In the two-subject setting, its LSD and High-LSD are very close to the best-performing baseline, while in the five-subject setting EpiTwin is best on all four metrics: RMSE, PCC, LSD, and High-LSD. These results indicate that the proposed model maintains a stable performance advantage even when subject heterogeneity and implantation diversity are introduced.

Table 3. **Performance on the Two-Subject Dataset (2 Patients).** Mean performance over 5 runs using subject-stratified sample-level splits; standard deviations are reported in Appendix D. Best results are **bolded**, second best are underlined.

Method	Log-Spectrogram		Freq-Domain	
	RMSE ↓	PCC ↑	LSD ↓	High-LSD ↓
Bi-LSTM	2.4148	0.9169	1.4793	1.3705
EEGNet	18.3232	0.4347	9.1342	11.2324
DGCNN	2.5665	0.9307	1.7556	1.8792
Brant	2.9449	0.9057	2.2549	2.1589
BrainBERT	2.1483	0.9546	1.4598	1.4021
LaBraM	<u>0.5922</u>	<u>0.9906</u>	0.6984	0.6955
EpiTwin	0.5731	0.9928	<u>0.6986</u>	<u>0.6988</u>

Table 4. **Performance on the Multi-Subject Dataset (5 Patients).** Mean performance over 5 runs using subject-stratified sample-level splits; standard deviations are reported in Appendix D. Best results are **bolded**, second best are underlined.

Method	Log-Spectrogram		Freq-Domain	
	RMSE ↓	PCC ↑	LSD ↓	High-LSD ↓
Bi-LSTM	2.0997	0.9531	1.2521	1.2169
EEGNet	14.6300	0.4270	8.9992	10.1384
DGCNN	2.1791	0.9612	1.5571	1.3552
Brant	2.4199	0.9179	1.7190	1.7908
BrainBERT	2.1505	0.9546	1.4623	1.4039
LaBraM	<u>0.6080</u>	<u>0.9895</u>	<u>0.6131</u>	<u>0.5928</u>
EpiTwin	0.5773	0.9896	0.6064	0.5753

To further rule out the potential influence of temporal auto-correlation and evaluate generalization to unseen subjects, we additionally conduct two stricter evaluation protocols. First, we adopt a temporally contiguous split, where the first 80% of the recordings are used for training and the last 20% are used for testing without shuffling. Second, we adopt a strict leave-one-subject-out (LOSO) protocol, where the model is trained on four subjects and tested on one entirely unseen subject.

As shown in Table 5, EpiTwin achieves an RMSE of 0.504 under the temporally contiguous split, compared with 0.492 under the original random split, corresponding to only a modest 2.4% degradation. This suggests that temporal auto-correlation has limited impact on the reported reconstruction performance. Moreover, EpiTwin achieves the best or tied-best performance across all metrics under this temporally stricter setting, with the lowest RMSE, LSD, and High-LSD, and a PCC comparable to the strongest baseline.

Under the more challenging LOSO protocol, EpiTwin remains the best-performing method across all evaluation metrics, achieving an RMSE of 0.956, PCC of 0.992, LSD of 0.870, and High-LSD of 0.825, as shown in Table 6. Com-

Table 5. Reconstruction performance under the temporally contiguous split. The first 80% of recordings are used for training and the last 20% for testing without shuffling.

Method	RMSE ↓	PCC ↑	LSD ↓	High-LSD ↓
Bi-LSTM	0.617	0.993	0.586	0.564
EEGNet	1.040	0.991	0.934	0.905
DGCNN	0.875	0.990	0.806	0.800
Brant	1.765	0.934	1.427	1.675
BrainBERT	1.335	0.974	1.142	1.222
LaBraM	0.556	0.989	0.517	0.495
EpiTwin	0.504	0.993	0.477	0.462

Table 6. Reconstruction performance under the strict leave-one-subject-out protocol. Models are trained on four subjects and tested on one unseen subject.

Method	RMSE ↓	PCC ↑	LSD ↓	High-LSD ↓
Bi-LSTM	1.389	0.977	1.243	1.197
EEGNet	12.627	0.100	11.885	8.736
DGCNN	1.894	0.949	1.787	1.883
Brant	1.704	0.935	1.416	1.690
BrainBERT	2.215	0.953	1.515	1.489
LaBraM	1.253	0.990	1.159	1.129
EpiTwin	0.956	0.992	0.870	0.825

pared with the strongest baseline LaBraM, EpiTwin reduces RMSE from 1.253 to 0.956 and LSD from 1.159 to 0.870. These results further demonstrate that EpiTwin generalizes robustly not only across heterogeneous multi-subject recordings, but also to completely unseen subjects.

5. Conclusion

We have proposed **EpiTwin**, a spatiotemporal graph transformer framework for high-fidelity sEEG signal reconstruction under sparse and irregular electrode coverage. By incorporating geometric, topological, and anatomical priors into masked reconstruction and attention modeling, EpiTwin achieves superior reconstruction accuracy and improved spectral fidelity. Experiments on clinical sEEG data show that EpiTwin consistently outperforms both conventional models and recent brain foundation models, while maintaining leading performance under increased inter-subject variability.

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Impact Statement

This paper presents methodological research aimed at advancing machine learning techniques for neural signal reconstruction. The study uses retrospective, de-identified clinical sEEG data and does not involve any intervention or clinical decision-making. All data collection and usage protocols were reviewed and approved by the relevant institutional ethics committee.

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A. STFT-Based Spectral Representation

This section provides the explicit formulation and parameterization of the Short-Time Fourier Transform (STFT) used for dynamic feature extraction.

Given a windowed sEEG signal $\mathbf{x}_k^{(i)} \in \mathbb{R}^W$ from electrode i , its STFT is computed as:

$$\mathbf{Z}_k^{(i)}[f, m] = \sum_{n=0}^{L_w-1} \mathbf{x}_k^{(i)}[n + mL_h] \cdot w[n] \cdot e^{-j \frac{2\pi f n}{N_{fft}}}. \quad (9)$$

$w[n]$ denotes the analysis window function (e.g., Hann window) of length L_w , L_h is the hop length (frame shift), and N_{fft} is the number of Fast Fourier Transform (FFT) points. The indices $f \in [0, F-1]$ and $m \in [0, M-1]$ denote the frequency bin and time frame, respectively. This results in

$$F = \frac{N_{fft}}{2} + 1, \quad M = \left\lfloor \frac{W - L_w}{L_h} \right\rfloor + 1, \quad (10)$$

frequency bins and time frames. The log-amplitude spectrogram is then obtained as $\mathbf{S}_k = \log(|\mathbf{Z}_k|)$.

B. Implementation Details

B.1. Hybrid Spatial Positional Encoding (HSPE)

This subsection provides implementation details and hyperparameter choices for the Hybrid Spatial Positional Encoding (HSPE) introduced in the main text. All formulations strictly follow the definitions in Section 3.1.

Geometric positional encoding. For the 3D sinusoidal positional encoding, the total feature dimension d is divided across the three spatial axes (x, y, z) . When d is not divisible by 3, we allocate $\lfloor d/3 \rfloor$ dimensions to each axis and assign the remaining dimension to the x -axis. The scale parameter is fixed to $\Omega = 10,000$, following the standard sinusoidal positional encoding convention. This encoding is parameter-free and remains constant during training and inference.

Laplacian eigenvector selection. For topological structure encoding, we discard the trivial zero eigenvector of the normalized graph Laplacian and select the first J smallest non-zero eigenvectors. The choice of J controls the resolution of global structural information and is treated as a hyperparameter. The eigenvectors are computed once prior to training and remain fixed throughout training and inference.

MLP architectures in HSPE. All projection modules used in HSPE, including the tissue composition projection $\mathbf{E}_M(\cdot)$, the Laplacian-based topological projection $\text{MLP}_t(\cdot)$, and the anatomical fusion projection $\text{MLP}_p(\cdot)$, are implemented as single-layer linear mappings without hidden layers. This design ensures minimal inductive bias while preserving computational efficiency.

Region embedding. The anatomical region embedding $\mathbf{E}_R(\cdot)$ is implemented as a learnable lookup table. Its embedding dimension is set to match the model hidden dimension d , ensuring compatibility with other static prior components.

B.2. CNN-based Dynamic Feature Encoder

The dynamic feature encoder operates on the log-amplitude spectrogram of each electrode independently and is implemented as a multi-scale convolutional network. Three parallel convolutional branches are employed to capture spectral patterns at fine, mid, and coarse resolutions. Each branch consists of stacked 3×3 convolutional layers followed by Batch Normalization and GELU activation, with strided convolutions used in the mid- and coarse-scale branches to enlarge the receptive field. The resulting feature maps are concatenated along the channel dimension, fused via additional 3×3 convolutions, and aggregated using adaptive average pooling. The final representation is projected to the model hidden dimension d via a linear layer.

B.3. Spatial Modeling and Graph Transformer Details

This section provides implementation details for the spatial graph construction, Geometry–Functional Biased Attention (GFBA), and the progressive decoding module described in Section 3.2.

Masked feature usage. In the masked graph construction, the shared query vector $\mathbf{q}$ used to replace target node features is identical to the query embedding used as input to the Transformer backbone. This design ensures strict consistency between topology inference and representation learning, and prevents information leakage from target nodes during both training and inference.

Data-driven functional similarity. Cosine similarity is adopted as the data-driven functional similarity measure when computing the adjacency matrix. This choice provides scale-invariant comparison between node features and is empirically stable for sEEG representations.

Spatial decay parameter. The bandwidth parameter σ of the Gaussian spatial decay kernel is set proportional to the median pairwise Euclidean distance between electrode coordinates within each subject. This adaptive scaling ensures robustness to variations in electrode density and implantation geometry across subjects.

Neighborhood sparsification. The hard spatial constraint is implemented using a k -nearest neighbor search in 3D Euclidean space. We fix $k = 8$ for all experiments unless otherwise stated, which yields a sparse yet sufficiently connected graph while reducing the computational complexity from $O(N^2)$ to $O(Nk)$.

Bias construction. The geometric bias and functional bias in GFBA are computed independently for each attention head and each Transformer layer. Both biases are generated by head-specific multilayer perceptrons (MLPs) that map a scalar edge attribute (Euclidean distance or adjacency weight) to a scalar bias value added to the attention logits.

Bias MLP architecture. Each bias MLP consists of two fully connected layers with ReLU activation. No normalization layers are used. Independent parameters are employed across attention heads to enable head-specific modulation of attention patterns.

Transformer configuration. The graph transformer backbone comprises $L = 20$ stacked Transformer blocks with $H = 8$ attention heads per block. All blocks follow a Pre-LN residual architecture. Dropout is applied only within the feed-forward sublayers.

Head aggregation strategy. For interaction analysis, attention weights are extracted from the final Transformer block, where higher-order and long-range dependencies are most fully integrated. The attention matrices from all heads are averaged to obtain a single attention-derived interaction matrix.

Interpretation of interaction weights. The resulting interaction matrix reflects the relative contribution of source nodes to target node reconstruction under the learned model dynamics. These weights are not optimized with explicit supervision and are intended solely for post-hoc qualitative analysis and visualization, rather than as a validated measure of physiological connectivity.

Decoder initialization. The final Transformer embeddings are linearly projected and reshaped into an initial feature map $\mathbf{f}^{(0)} \in \mathbb{R}^{C_D \times F_0 \times T}$, where C_D denotes the number of feature channels, F_0 is the initial frequency resolution, and T is the number of time frames, which exactly matches the temporal resolution of the input spectrogram.

Frequency-only upsampling. The decoder employs anisotropic transposed convolutions that progressively upsample only along the frequency dimension while keeping the temporal dimension fixed. This design preserves temporal precision and prevents smearing of transient spectral events.

Output projection. A final 1×1 convolution maps the decoder feature map to the reconstructed log-amplitude spectrogram $\hat{\mathbf{s}}_i \in \mathbb{R}^{F \times M}$. Specific kernel sizes, channel widths, and stride configurations are reported in the implementation table.

B.4. Decoder Architecture Details

This section provides the full implementation details of the Frequency-Exclusive Progressive Decoder used in EpiTwin.

Initial Feature Map. The decoder starts from the latent representation $\mathbf{z}^{(L)}$, which is projected through a fully connected layer $\mathbf{W}_D$ and reshaped into an initial feature map

$$\mathbf{f}^{(0)} \in \mathbb{R}^{C_D \times F_0 \times T},$$

where C_D denotes the number of decoder channels, F_0 is the initial spectral resolution, and T is the temporal resolution. In our implementation, we set $F_0 = 8$ and $T = 16$, such that the temporal resolution exactly matches that of the input representation.

Frequency-Exclusive Progressive Upsampling. To progressively recover spectral resolution, the decoder employs a three-stage anisotropic upsampling strategy. Each stage uses a two-dimensional transposed convolution configured to upsample exclusively along the frequency dimension, while keeping the temporal dimension unchanged. Specifically, each transposed convolution uses a kernel size of $(k_f, 1)$, a stride of $(2, 1)$, and padding of $(p_f, 0)$. This configuration results in frequency resolutions of $8 \rightarrow 16 \rightarrow 32 \rightarrow 64$ across the three stages, with the temporal resolution remaining fixed at $T = 16$ throughout decoding.

Unless otherwise specified, we use the same transposed convolution configuration for all upsampling stages, with $k_f = 4$ and $p_f = 1$ in our implementation.

Output Projection. After the final upsampling stage, a 1×1 convolution is applied to project the resulting feature map of size $C_D \times 64 \times 16$ to the target output $\hat{\mathbf{s}}_i$.

B.5. Two-Stage Training and Adversarial Optimization

This section provides implementation details and additional explanations for the two-stage training strategy used for sEEG spectrogram reconstruction.

Role of L1 initialization. The L1 reconstruction loss provides stable gradients and encourages the model to learn the correct coarse-grained spectral morphology. However, for high-resolution spectrogram generation, pure L1 optimization is known to induce regression-to-the-mean behavior, leading to over-smoothed reconstructions that lack fine-grained details.

Optimization details. During Phase 1, the generator is optimized using Adam with fixed learning rate and default momentum parameters. Training proceeds until convergence, and the resulting parameters are used to initialize the adversarial refinement phase.

Multi-scale discriminator. The multi-scale discriminator consists of two PatchGAN-based sub-discriminators operating at different resolutions, enabling the assessment of both local textures and global spectral structure.

Spectral normalization. Spectral normalization is applied to all convolutional layers in the discriminator to stabilize adversarial training and prevent exploding gradients.

Perceptual loss. Perceptual loss $\mathcal{L}_p$ is computed as the L1 distance between intermediate feature representations extracted from all convolutional layers of the discriminator except the final output layer. This loss promotes the recovery of high-level textural patterns that are not well captured by pixel-wise objectives.

Loss weighting and alternating optimization. In the adversarial refinement phase, the reconstruction loss is down-weighted to $\lambda_r = 0.5$, while the adversarial and perceptual losses are weighted by $\lambda_a = 1.0$ and $\lambda_p = 0.5$, respectively. The generator and discriminator are trained in an alternating fashion with $N_D = 1$ and $N_G = 1$, which provides stable convergence in practice.

C. Baselines and Evaluation Metrics

C.1. Baseline Models

We benchmark EpiTwin against two categories of baseline methods.

Conventional Reconstruction Models. Bi-LSTM is implemented as a bidirectional recurrent neural network for multichannel sEEG reconstruction, serving as a strong temporal modeling baseline. EEGNet is a compact convolutional architecture originally proposed for EEG decoding and is adapted here for signal reconstruction. DGCNN represents graph-based approaches that model spatial relationships between channels using dynamically constructed graphs.

Brain Foundation Models. We include representative foundation models for intracranial neural signals, including BrainBERT, Brant, and LaBraM. BrainBERT employs masked modeling with Transformer encoders for neural time series. Brant is a foundation model designed for large-scale intracranial recordings. LaBraM adopts a patch-based Transformer architecture for learning generic EEG representations.

All baseline models are retrained from scratch using the same STFT-based inputs, series-aware masking strategy, and evaluation splits as EpiTwin. For foundation models, default architectural configurations are retained, and input layers are adapted to match the spectrogram representation used in this work. No pretrained weights are used.

C.2. Evaluation Metrics

We evaluate reconstruction quality using log-spectrogram and frequency-domain metrics, as detailed below.

Root Mean Squared Error (RMSE). RMSE measures the average magnitude of reconstruction error and is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (11)$$

where y_i and $\hat{y}_i$ denote the ground-truth and reconstructed signals, respectively. The summation is taken over all elements across samples and signal dimensions.

Pearson Correlation Coefficient (PCC). PCC quantifies the linear correlation between reconstructed and ground-truth signals:

$$\text{PCC} = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}} \quad (12)$$

where $\bar{y}$ and $\bar{\hat{y}}$ denote the mean of the ground-truth and reconstructed signals, respectively. PCC is computed on the log-spectrogram signals after flattening all dimensions.

Log-Spectral Distance (LSD). LSD evaluates spectral distortion between reconstructed and ground-truth signals. Let $R_{m,k}$ and $S_{m,k}$ denote the magnitude spectra of the ground-truth and reconstructed signals for sample m and frequency bin k , respectively. LSD is defined as

$$\text{LSD} = \frac{1}{M} \sum_{m=1}^M \sqrt{\frac{1}{K} \sum_{k=1}^K \left(20 \log_{10} \left(\frac{S_{m,k}}{R_{m,k}} + \varepsilon \right) \right)^2}, \quad (13)$$

where M is the number of samples, K is the number of frequency bins, and ε is a small constant used for numerical stability.

High-Frequency Log-Spectral Distance (High-LSD). To assess spectral fidelity in higher frequency bands, we compute LSD over the upper half of the frequency spectrum. Let $H \subset \{1, \dots, K\}$ denote the set of high-frequency bins, defined as the latter half of the frequency axis. High-LSD is given by

$$\text{High-LSD} = \frac{1}{M} \sum_{m=1}^M \sqrt{\frac{1}{|H|} \sum_{k \in H} \left(20 \log_{10} \left(\frac{S_{m,k}}{R_{m,k}} + \varepsilon \right) \right)^2}. \quad (14)$$

Table 7. **Performance on the Two-Subject Dataset (2 Patients).** We adopt subject-stratified *sample-level* splits, where all two subjects appear in the training/validation/test sets, and report the mean $\pm$ standard deviation over 5 independent runs. Best results are **bolded**, second best are underlined.

Method	Log-Spectrogram		Freq-Domain	
	RMSE $\downarrow$	PCC $\uparrow$	LSD $\downarrow$	High-LSD $\downarrow$
Bi-LSTM	2.4148 ± 0.0123	0.9169 ± 0.0045	1.4793 ± 0.0187	1.3705 ± 0.0201
EEGNet	18.3232 ± 0.2154	0.4347 ± 0.0128	9.1342 ± 0.1732	11.2324 ± 0.2546
DGCNN	2.5665 ± 0.0216	0.9307 ± 0.0039	1.7556 ± 0.0194	1.8792 ± 0.0221
Brant	2.9449 ± 0.0342	0.9057 ± 0.0067	2.2549 ± 0.0283	2.1589 ± 0.0315
BrainBERT	2.1483 ± 0.0264	0.9546 ± 0.0032	1.4598 ± 0.0179	1.4021 ± 0.0196
LaBraM	<u>0.5922 ± 0.0061</u>	<u>0.9906 ± 0.0008</u>	0.6984 ± 0.0092	0.6955 ± 0.0087
EpiTwin	<u>0.5731 ± 0.0054</u>	<u>0.9928 ± 0.0006</u>	<u>0.6986 ± 0.0079</u>	<u>0.6988 ± 0.0081</u>

All metrics are computed on held-out test samples and reported as averages over multiple independent runs.

D. Additional Qualitative Results

This section provides complementary qualitative and extended quantitative results that support the main findings reported in the paper. All results are evaluated under the same experimental protocol described in Section 4.1.

Holistic Performance Comparison. Figure 4 presents a normalized radar chart comparing EpiTwin with representative traditional and foundation models across multiple evaluation metrics. This holistic view highlights that EpiTwin achieves a more balanced trade-off between log-spectrogram fidelity and spectral realism, covering the largest normalized area among all methods.

Extended Ablation Analysis. Figure 5 visualizes the component-wise ablation results corresponding to Table 2 in the main text. Removing any individual component consistently degrades reconstruction quality, while the full EpiTwin model maintains superior performance across all evaluated metrics, further validating the necessity of each design choice.

Additional Multi-Subject Quantitative Results. Tables 7 and 8 report extended quantitative results under the two-subject and five-subject settings, respectively, including mean and standard deviation over multiple runs. These results complement the main quantitative comparisons by providing full statistical variability and show that EpiTwin maintains the best log-spectrogram reconstruction performance, while achieving competitive or best frequency-domain performance under increased inter-subject variability.

Attention-Based Interaction Visualization. Figure 6 illustrates the attention weight matrix learned by the graph transformer during reconstruction. This visualization reflects reconstruction-driven interaction patterns across sEEG channels and serves as an interpretable by-product of the proposed geometry-functional biased attention mechanism, rather than a separately supervised connectivity estimate.

Frequency-Time Error Reduction Analysis. Figure 7 shows the mean reconstruction error reduction aggregated across bipolar channels. The heatmap highlights frequency-time regions where EpiTwin achieves larger improvements, providing additional insight into its ability to preserve fine-grained spectral structure beyond global summary metrics.

Qualitative Waveform Reconstruction Examples. Finally, Figure 8 presents representative waveform reconstruction examples. Across different channels and temporal segments, the reconstructed signals closely follow the ground-truth waveforms, illustrating that the proposed model preserves both amplitude dynamics and temporal structure at the signal level.

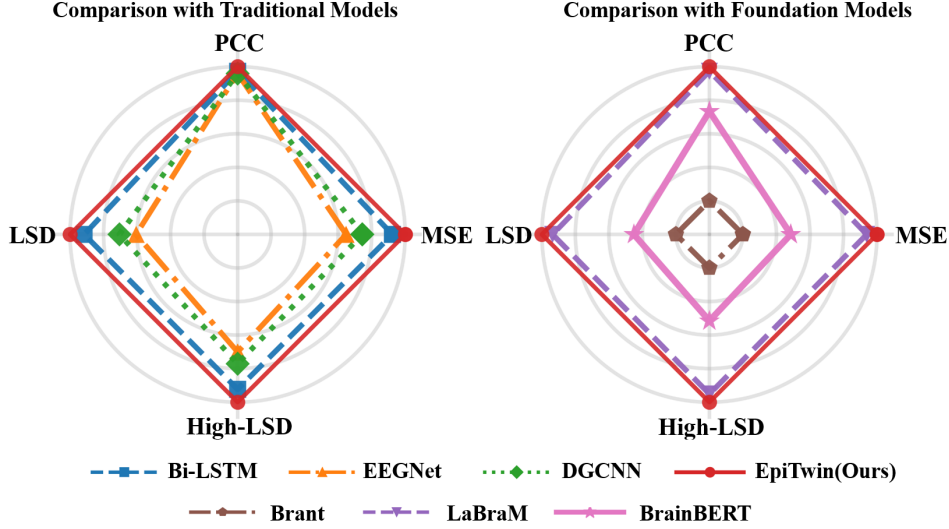


Figure 4. **Holistic Performance Comparison.** Normalized metrics showing that EpiTwin (red) achieves the best balance between log-spectrogram fidelity and spectral realism, covering the largest area among all methods.

Table 8. **Performance on the Multi-Subject Dataset (5 Patients).** We adopt subject-stratified *sample-level* splits, where all five subjects appear in the training/validation/test sets, and report the mean $\pm$ standard deviation over 5 independent runs. Best results are **bolded**, second best are underlined.

Method	Log-Spectrogram		Freq-Domain	
	RMSE $\downarrow$	PCC $\uparrow$	LSD $\downarrow$	High-LSD $\downarrow$
Bi-LSTM	2.0997 ± 0.0021	0.9531 ± 0.0023	1.2521 ± 0.0039	1.2169 ± 0.0034
EEGNet	14.6300 ± 0.0913	0.4270 ± 0.0250	8.9992 ± 0.1142	10.1384 ± 0.1324
DGCNN	2.1791 ± 0.0019	0.9612 ± 0.0040	1.5571 ± 0.0157	1.3552 ± 0.0023
Brant	2.4199 ± 0.0154	0.9179 ± 0.0082	1.7190 ± 0.0184	1.7908 ± 0.0102
BrainBERT	2.1505 ± 0.0223	0.9546 ± 0.0051	1.4623 ± 0.0155	1.4039 ± 0.0146
LaBraM	<u>0.6080 ± 0.0060</u>	<u>0.9895 ± 0.0015</u>	<u>0.6131 ± 0.0060</u>	<u>0.5928 ± 0.0055</u>
EpiTwin	0.5773 ± 0.0055	0.9896 ± 0.0012	0.6064 ± 0.0060	0.5753 ± 0.0050

E. Dataset Construction and Experimental Settings

E.1. Raw Dataset and Metadata

The raw sEEG dataset consists of recordings from multiple patients with drug-resistant epilepsy. For each patient, sEEG signals are transformed into bipolar differential signals and represented as log-amplitude STFT spectrograms. Each patient dataset includes the following components:

- Bipolar STFT features stored in `patient_XXX/bipolar_stft.h5`;
- Spatial coordinates of bipolar channels in `*_coords.npy`;
- Tissue composition metadata in `*_tissue.npy`;
- Anatomical region labels in `*_regions.npy`.

These static metadata are fixed per patient and shared across all experimental settings.

E.2. Single-Patient Setting

In the single-patient setting, all samples are drawn from `patient_001`, resulting in 8,604 time-windowed samples. Samples are randomly split into training (60%), validation (20%), and test (20%) sets at the sample level.

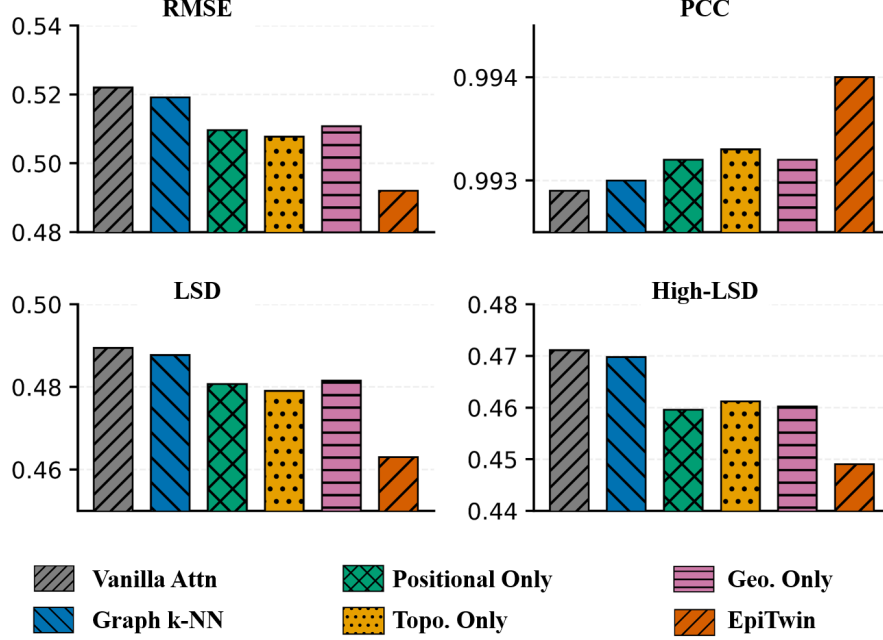


Figure 5. **Component-wise Ablation Analysis.** Visualization of reconstruction performance across different ablated variants, illustrating the consistent advantage of the full EpiTwin model over its component-removed counterparts.

E.3. Two-Patient Setting

The two-patient setting combines recordings from `patient_001` and `patient_002`, yielding a total of 13,811 samples (8,604 from `patient_001` and 5,207 from `patient_002`).

E.4. Five-Patient Subset Setting

To evaluate robustness under increased inter-subject heterogeneity while controlling for dataset scale, we construct a five-patient subset using recordings from `patient_001`, `patient_002`, `patient_003`, `patient_004`, and `patient_007`. From the full pool of 32,525 available samples across these patients, we randomly select 10,000 samples without replacement. Sampling probabilities are proportional to each patient’s original sample count, with a minimum of one sample per patient.

Channel Alignment Across Subjects. Since the number of bipolar channels varies substantially across patients, all samples in this setting are aligned to a common channel dimension of 81, corresponding to the minimum number of available channels among the included subjects. This controlled construction isolates the effect of increased subject diversity while minimizing confounding factors arising from dataset size and channel dimensionality.

For each patient, we rank bipolar channels using a composite importance score and retain the top 81 channels. Let $c \in \{1, \dots, C_p\}$ index the bipolar channels of patient p . For channel c , we compute three normalized signal-quality measures: energy, signal-to-noise ratio (SNR), and variance,

$$E_c = \frac{\mathcal{E}_c}{\max_{c'} \mathcal{E}_{c'}}, \quad S_c = \frac{|\text{SNR}_c|}{\max_{c'} |\text{SNR}_{c'}|}, \quad V_c = \frac{\sigma_c^2}{\max_{c'} \sigma_{c'}^2}, \quad (15)$$

where $\mathcal{E}_c$ denotes the total signal energy of channel c , SNR_c is estimated as the ratio between the mean and standard deviation of the signal, and σ_c^2 denotes signal variance (Alotaiby et al., 2015). All quantities are normalized within each patient.

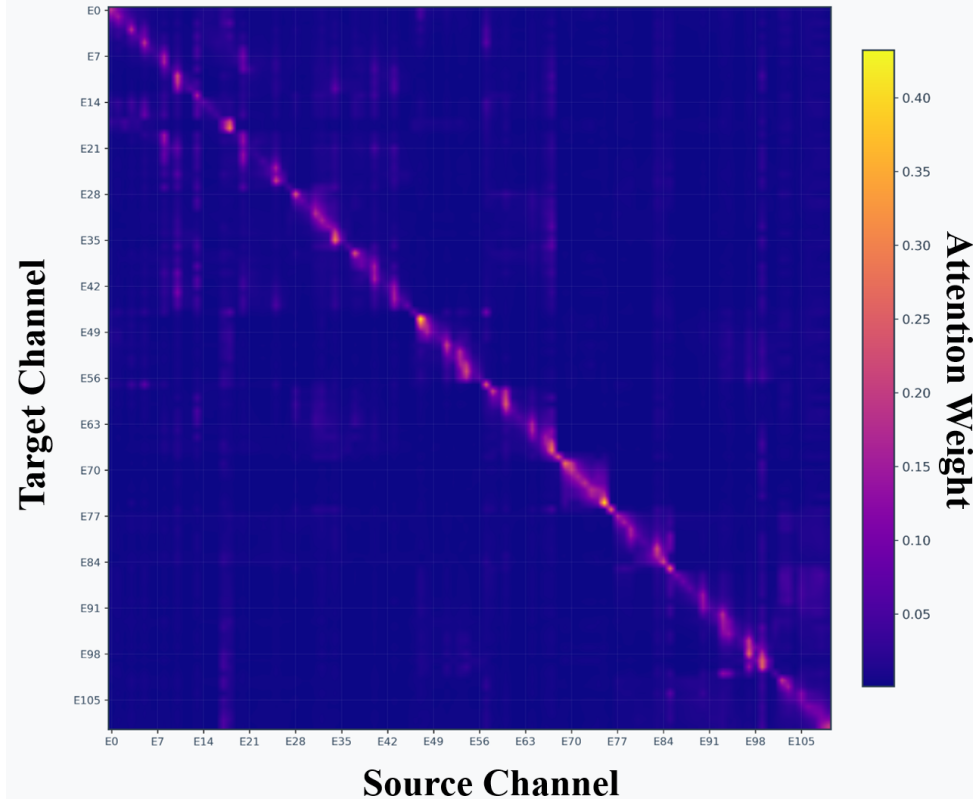


Figure 6. Learned attention weight matrix across sEEG recording channels during reconstruction.

Table 9. **Channel Alignment Summary Across Patients.** All patients are aligned to a common channel dimension of 81 for multi-subject experiments. Patients with fewer than 81 bipolar channels are excluded to avoid artificial padding.

Patient ID	Original Channels	Alignment Strategy	Final Channels
patient_001	110	Top-81 selection	81
patient_002	124	Top-81 selection	81
patient_003	81	Unchanged	81
patient_004	98	Top-81 selection	81
patient_007	103	Top-81 selection	81
patient_008	54	Excluded	–

The final composite importance score is defined as a weighted sum:

$$I_c = 0.4 E_c + 0.4 S_c + 0.2 V_c. \quad (16)$$

Channels are ranked in descending order of I_c , and the top 81 channels are retained for that patient.

Patients with fewer than 81 bipolar channels are excluded from the multi-subject setting to avoid artificial padding or extrapolation. This alignment strategy preserves the most informative channels for each subject while ensuring a uniform channel dimensionality across patients, enabling fair evaluation under heterogeneous implantation patterns. A summary of channel alignment decisions for each patient is provided in Table 9.

E.5. Patient 001 Dataset

We utilize stereotactic electroencephalography (sEEG) recordings from epilepsy patients for our experiments. The dataset consists of bipolar differential signals derived from monopolar electrode recordings, providing spatial information about electrode positions, tissue compositions, and anatomical brain regions.

The complete dataset includes 5 patients with variable numbers of electrodes (81–124 channels per patient), totaling 32,525

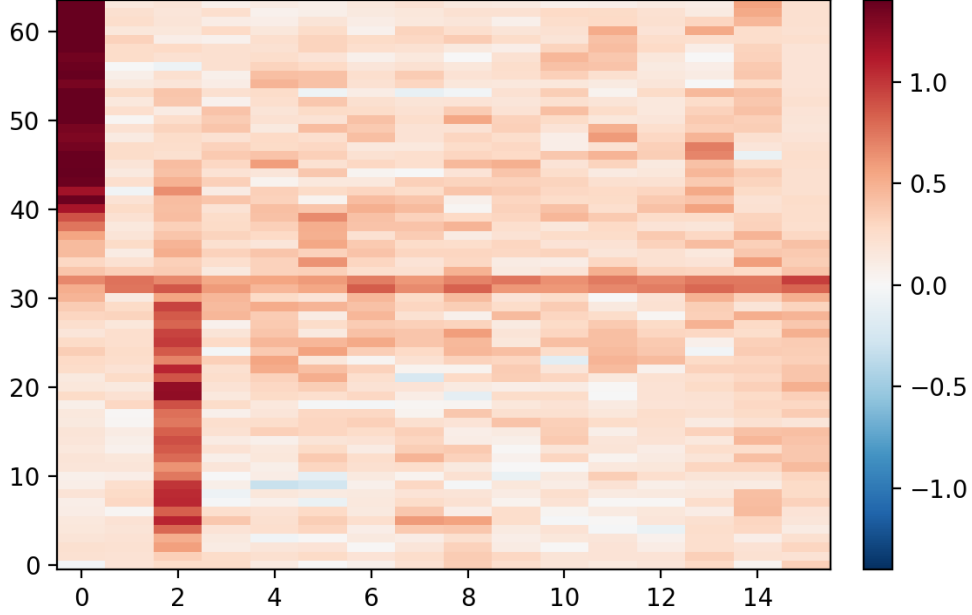


Figure 7. Mean Reduction Across Channels. Visualization of the average reconstruction error reduction across bipolar channels. Each row corresponds to a frequency bin and each column corresponds to a time frame. Values are averaged across samples and channels, highlighting frequency–time regions where EpiTwin achieves larger error reduction.

samples. In this document, we focus on the detailed description of Patient 001, which serves as the primary dataset for single-patient experiments.

E.6. Basic Information

Table 10. Basic characteristics of Patient 001 dataset

Property	Value
Patient ID	patient_001
Electrode Configuration	Bipolar differential
Number of Channels	110 bipolar pairs
Number of Samples	8,604
Sampling Rate	200 Hz
Window Length	10 seconds
Frequency Band	0.5–50 Hz
STFT Bins	64
Time Frames	16
Number of Brain Regions	16

E.6.1. ELECTRODE SERIES DISTRIBUTION

Patient 001 utilizes 7 electrode series, comprising 110 bipolar electrode pairs derived from adjacent electrode subtractions. Table 11 summarizes the electrode series distribution.

Table 11. Electrode series distribution for Patient 001

Series	Number of Bipolar Pairs	Electrode Range
A	17	A1-A18 $\rightarrow$ A1-A2, A2-A3, ..., A17-A18
B	17	B1-B18 $\rightarrow$ B1-B2, B2-B3, ..., B17-B18
C	17	C1-C18 $\rightarrow$ C1-C2, C2-C3, ..., C17-C18
D	17	D1-D18 $\rightarrow$ D1-D2, D2-D3, ..., D17-D18
E	17	E1-E18 $\rightarrow$ E1-E2, E2-E3, ..., E17-E18
F	17	F1-F18 $\rightarrow$ F1-F2, F2-F3, ..., F17-F18
G	8	G1-G9 $\rightarrow$ G1-G2, G2-G3, ..., G8-G9
Total	110	119 monopolar electrodes

E.6.2. BRAIN REGION DISTRIBUTION

The electrodes are distributed across 16 distinct anatomical regions in the left hemisphere, covering temporal, frontal, and occipital lobes. Table 12 provides the detailed distribution of electrodes across brain regions.

Table 12. Brain region distribution for Patient 001 electrodes

Region ID	Region Name	Electrode Count	Lobe
0	Hippocampus	9	Temporal (medial)
1	Unknown	7	–
2	White-Matter	10	Subcortical
3	ctx-lh-fusiform	1	Temporal
4	ctx-lh-inferiortemporal	5	Temporal
5	ctx-lh-lateraloccipital	2	Occipital
6	ctx-lh-lingual	7	Occipital
7	ctx-lh-middletemporal	4	Temporal
8	ctx-lh-parsorbitalis	3	Frontal (orbital)
9	ctx-lh-pericalcarine	2	Occipital
10	ctx-lh-rostralmiddlefrontal	4	Frontal
11	ctx-lh-superiortemporal	15	Temporal
12	Other Region 12	5	–
13	Other Region 13	8	–
14	Other Region 14	24	–
15	Other Region 15	4	–
Total	16 regions	110 electrodes	–

E.6.3. DATASET STATISTICS SUMMARY

Table 13. Summary statistics for Patient 001

Metric	Value
Total Recording Duration	$\approx$ 23.9 hours
Number of 10s Windows	8,604
Average Electrodes per Region	6.88
Temporal Lobe Coverage	35 electrodes (32%)
Frontal Lobe Coverage	8 electrodes (7%)
Occipital Lobe Coverage	11 electrodes (10%)
Medial/Subcortical Coverage	19 electrodes (17%)
Unknown/Other Regions	37 electrodes (34%)

Table 14. **Hyperparameter Sensitivity Analysis.** Performance comparison under different configurations. The optimal settings are highlighted in **bold**.

Hyperparam.	Value	Metrics			
		RMSE↓	PCC↑	MAE↓	LSD↓
Learning Rate	1×10^{-4}	0.516	0.991	0.387	0.475
	5×10^{-4}	0.497	0.993	0.372	0.464
	1×10^{-3}	0.490	0.994	0.369	0.461
	2×10^{-3}	0.494	0.993	0.371	0.463
	5×10^{-3}	0.533	0.987	0.400	0.486
Model Depth (L)	8	0.512	0.992	0.384	0.473
	12	0.500	0.993	0.375	0.466
	16	0.493	0.994	0.371	0.463
	20	0.492	0.994	0.369	0.461
	24	0.492	0.994	0.369	0.462
Hidden Dim. (d)	128	0.509	0.992	0.381	0.471
	192	0.500	0.993	0.374	0.466
	256	0.494	0.994	0.371	0.463
	384	0.491	0.994	0.369	0.462
	512	0.490	0.994	0.369	0.461

F. Hyperparameter Sensitivity

We assess the robustness of EpiTwin by analyzing the impact of three critical hyperparameters: Learning Rate, Model Depth, Hidden Dimension. The results are summarized in Table 14.

Learning Rate. Optimization stability is crucial for training graph transformers. We observe that performance peaks at a learning rate of 1×10^{-3} . Lower rates (1×10^{-4}) result in suboptimal convergence (RMSE 0.512), likely getting trapped in local minima. Conversely, excessively high rates (5×10^{-3}) destabilize the training process, causing loss oscillation and degrading reconstruction fidelity.

Model Depth (L). The network depth governs the receptive field size and the extent of information propagation. Performance improves consistently as depth increases from 8 to 16, indicating that deeper layers are necessary to capture long-range epileptic propagation pathways. However, performance saturates and slightly degrades at $L = 24$. This decline is attributed to the *over-smoothing* problem inherent in deep GNNs, where node representations tend to become indistinguishable as graph convolution layers stack up.

Hidden Dimension (d). This parameter controls the model’s representation capacity. A dimension of $d = 256$ strikes the optimal balance between expressivity and generalization. Smaller dimensions ($d = 128$) constrain the latent space, failing to encode complex high-frequency textures. Larger dimensions ($d = 512$), while expressive, introduce parameter redundancy and increase the risk of overfitting on the sparse sEEG dataset, leading to a slight drop in PCC.

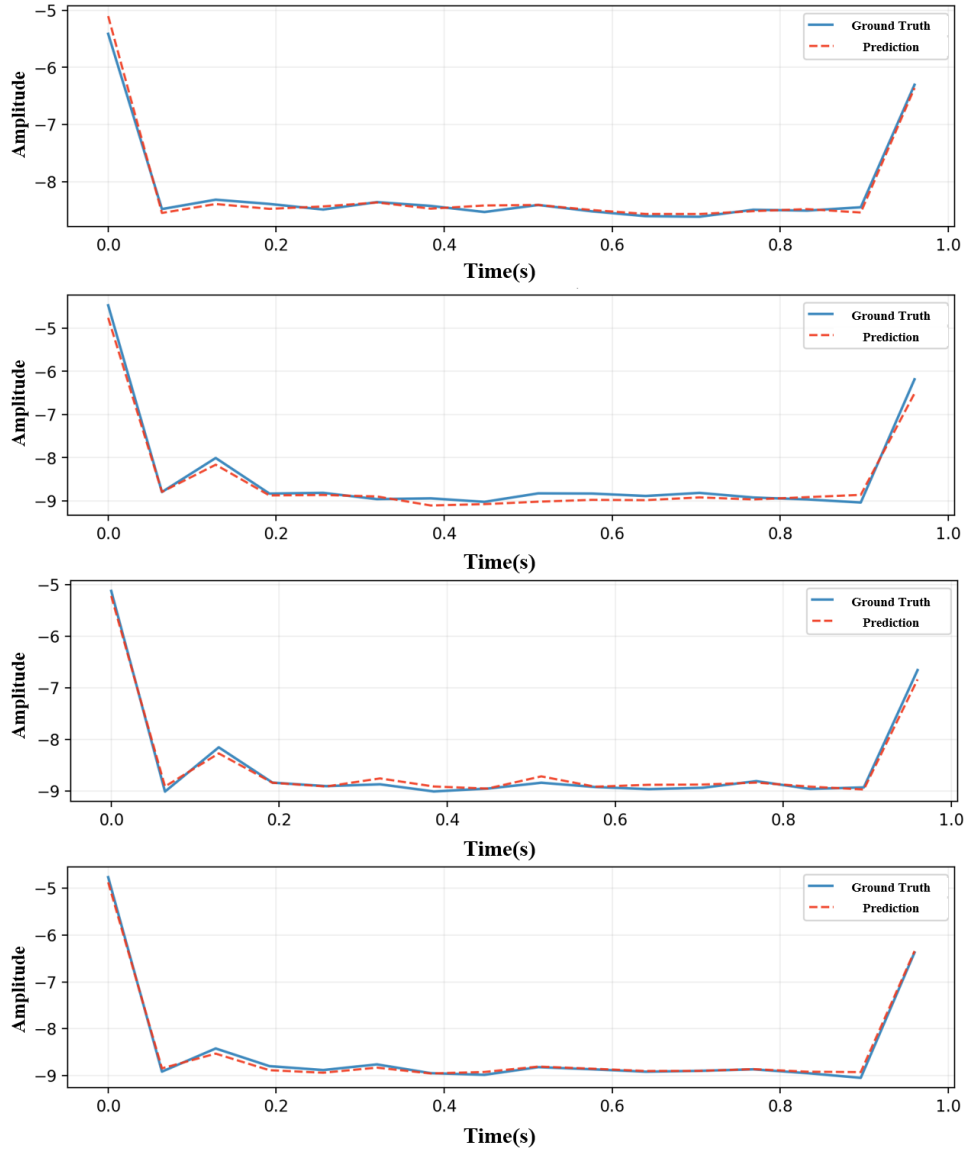


Figure 8. Qualitative waveform reconstruction examples on sEEG signals. Each subplot shows a representative channel segment, comparing the ground-truth waveform (solid blue) and the reconstructed waveform (dashed red) over time.